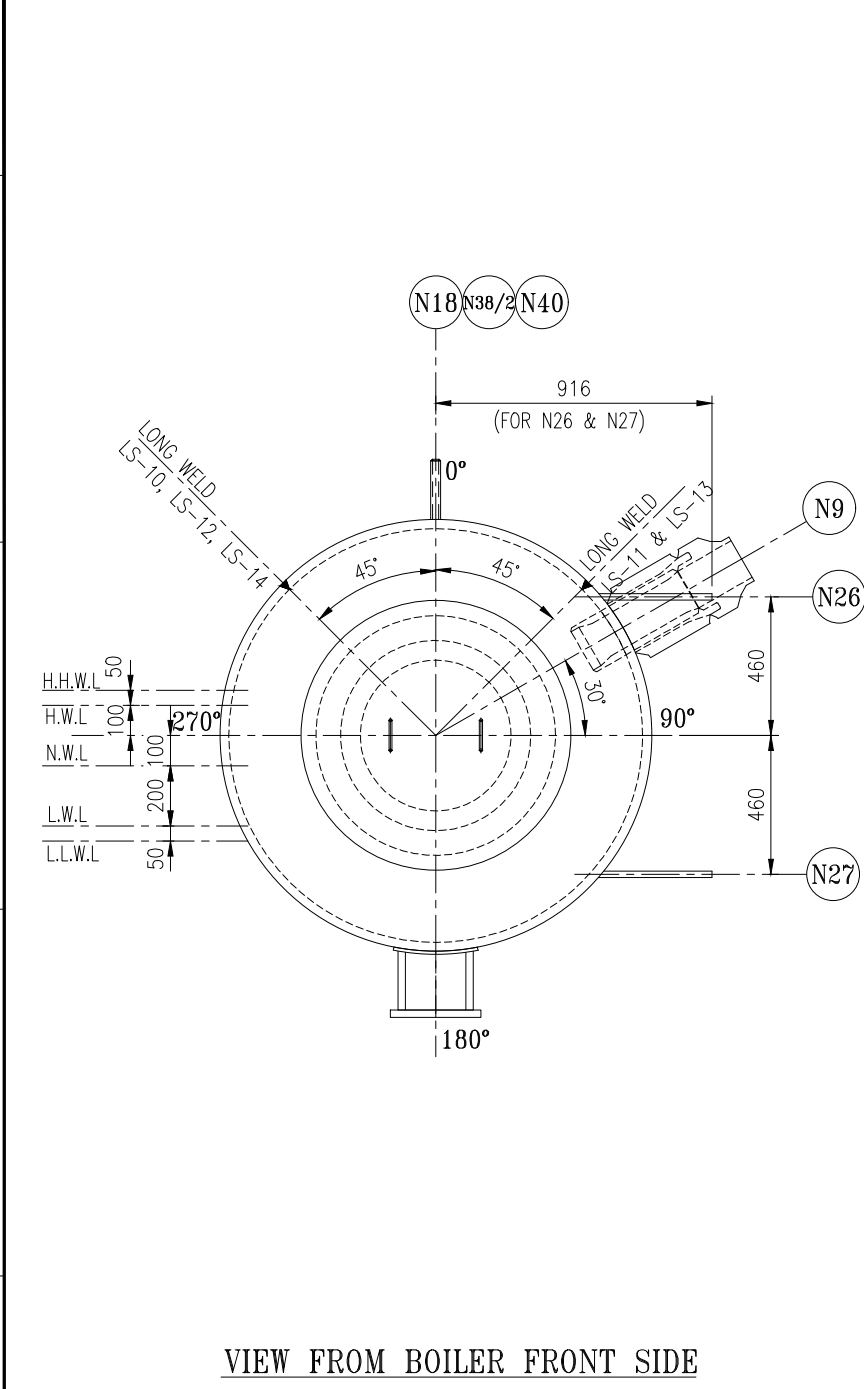
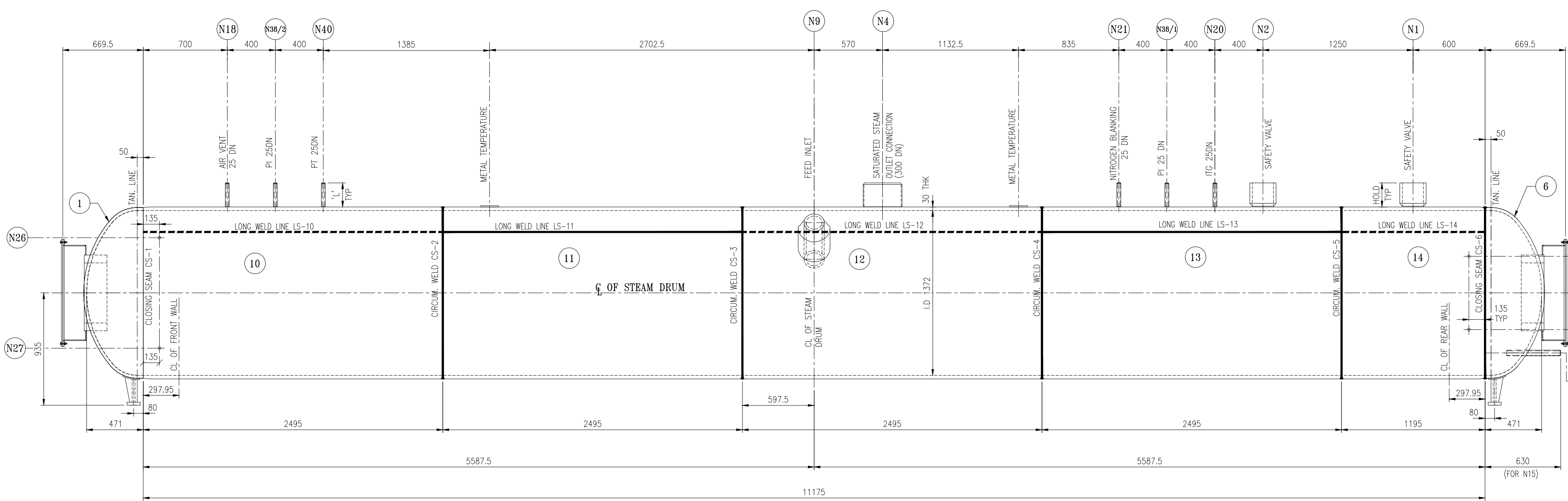


THIS IS A CAD FILE - DO NOT CORRECT IT MANUALLY

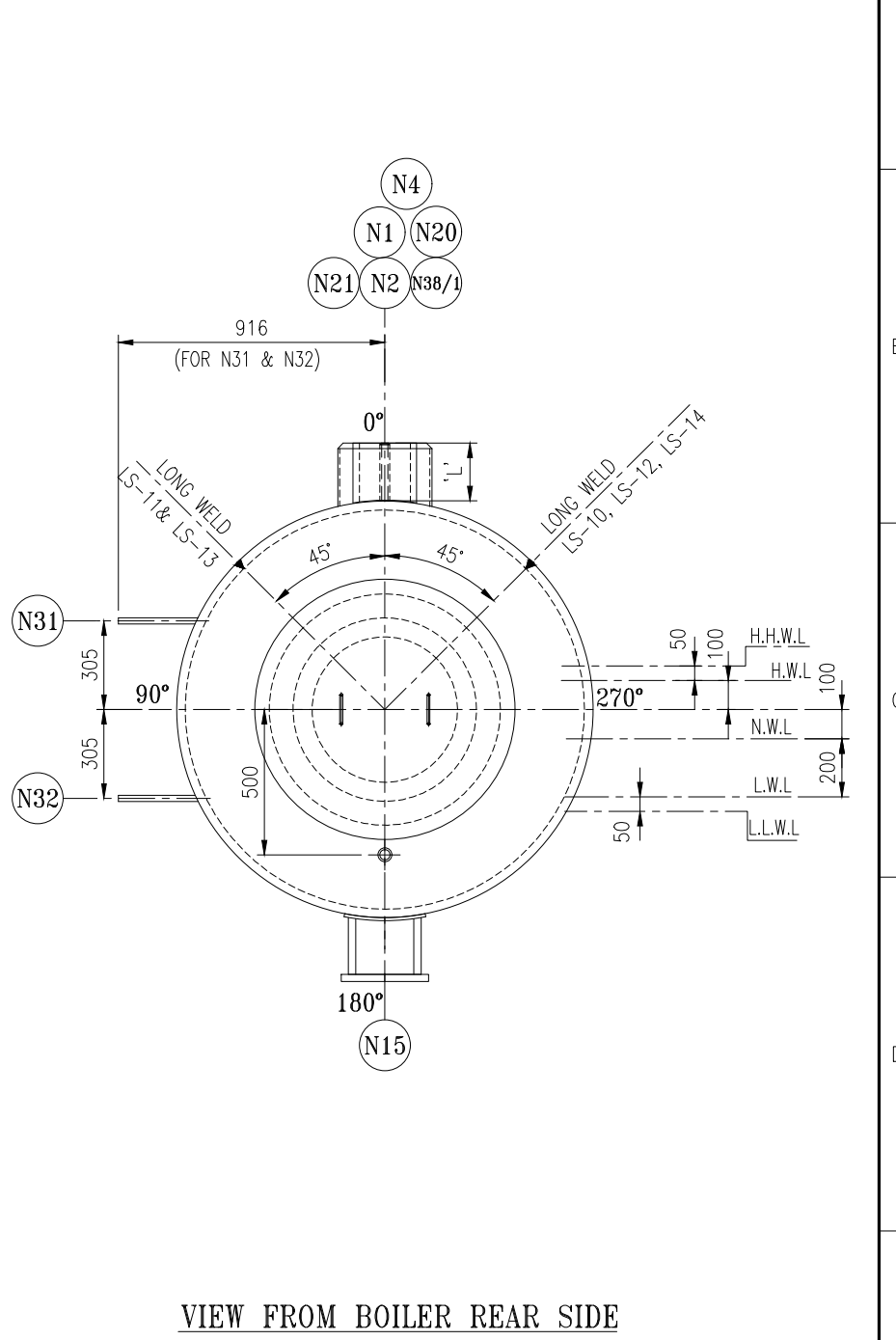
1000-Z00-0010101000	DOCUMENT ID	JB1179
	PROJECT ID	JB1179



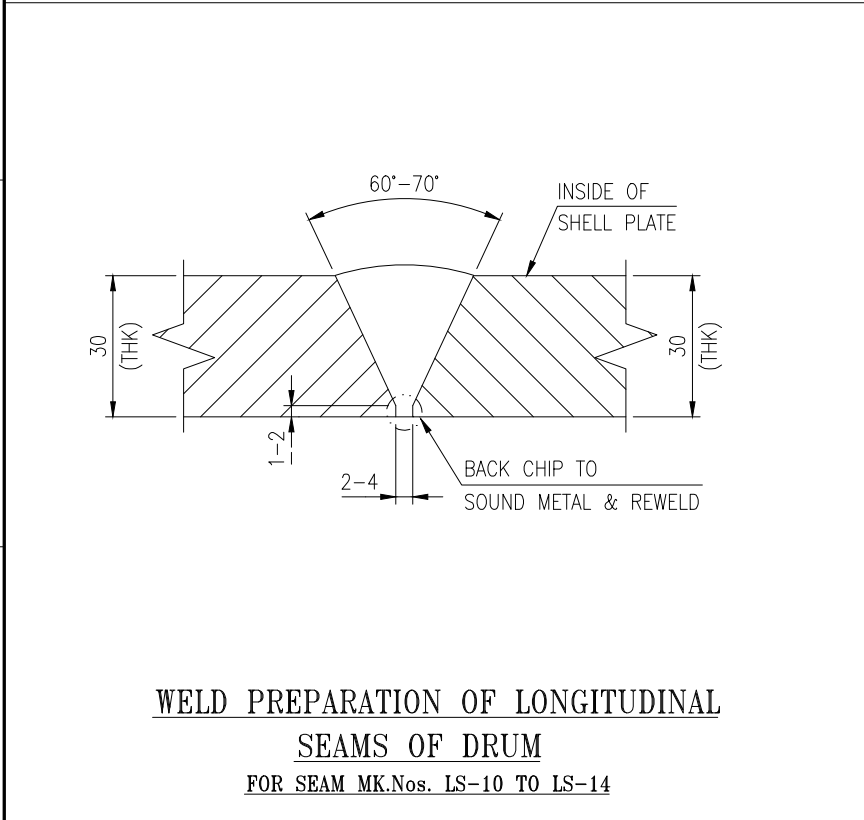
VIEW FROM BOILER FRONT SIDE



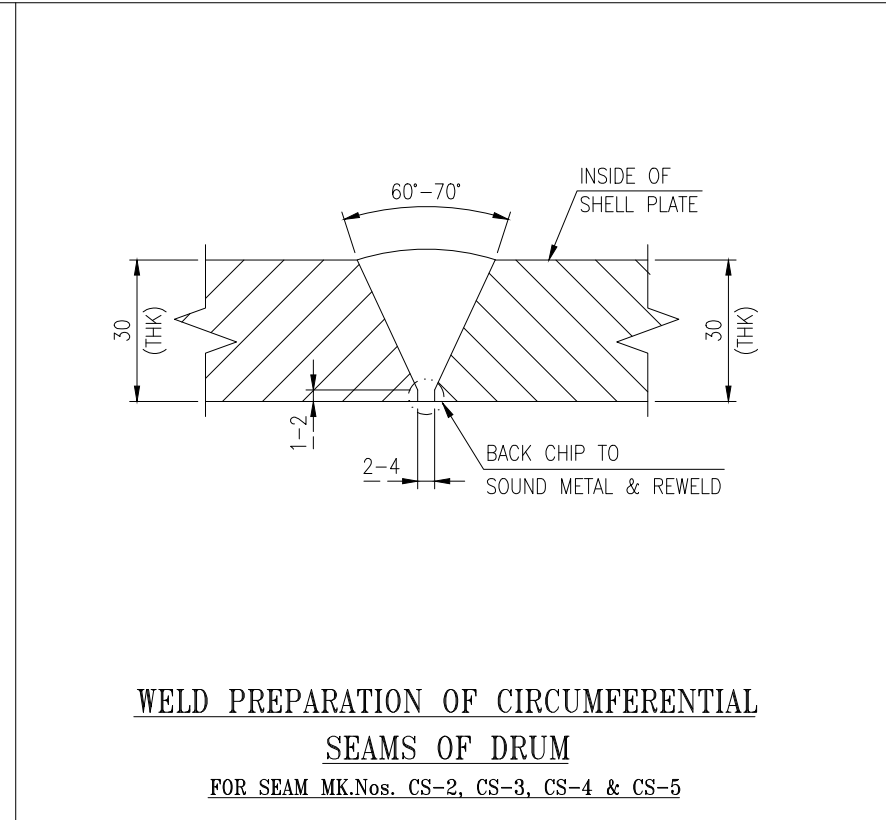
ARRANGEMENT OF STEAM DRUM
ELEVATION FROM R.H.S. OF BOILER



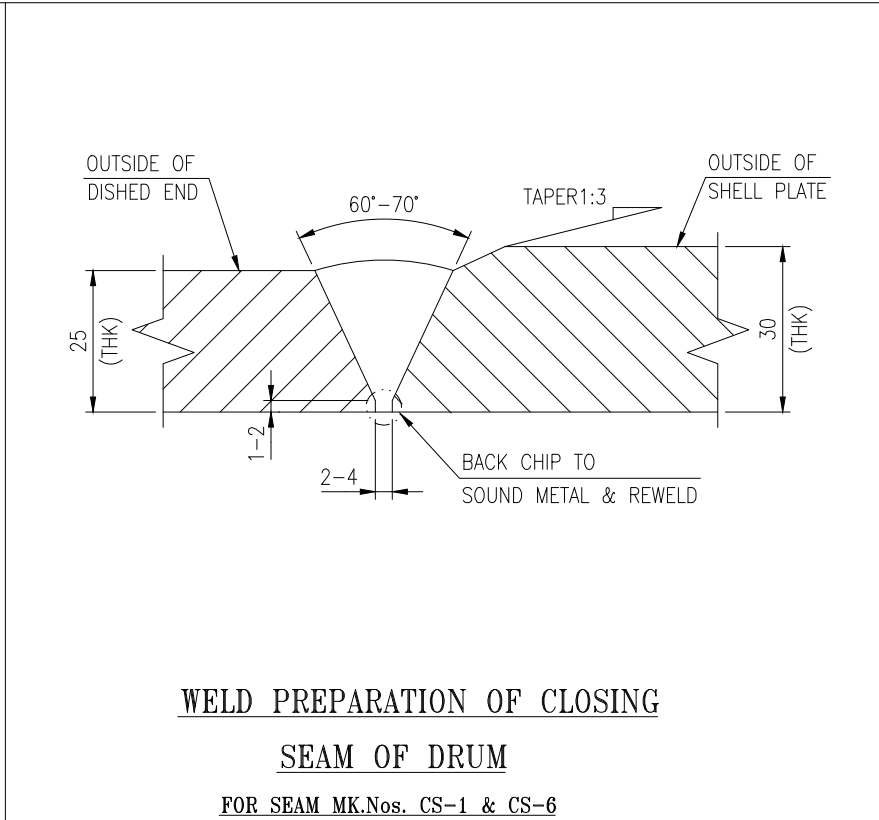
VIEW FROM BOILER REAR SIDE



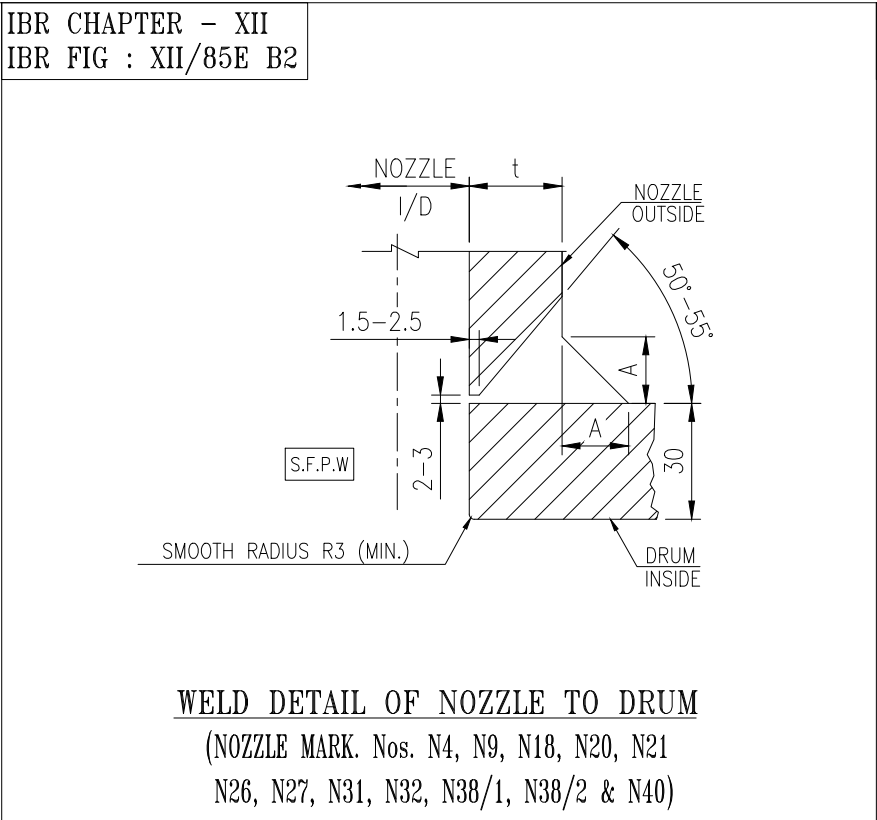
WELD PREPARATION OF LONGITUDINAL
SEAMS OF DRUM
FOR SEAM MK.Nos. LS-10 TO LS-14



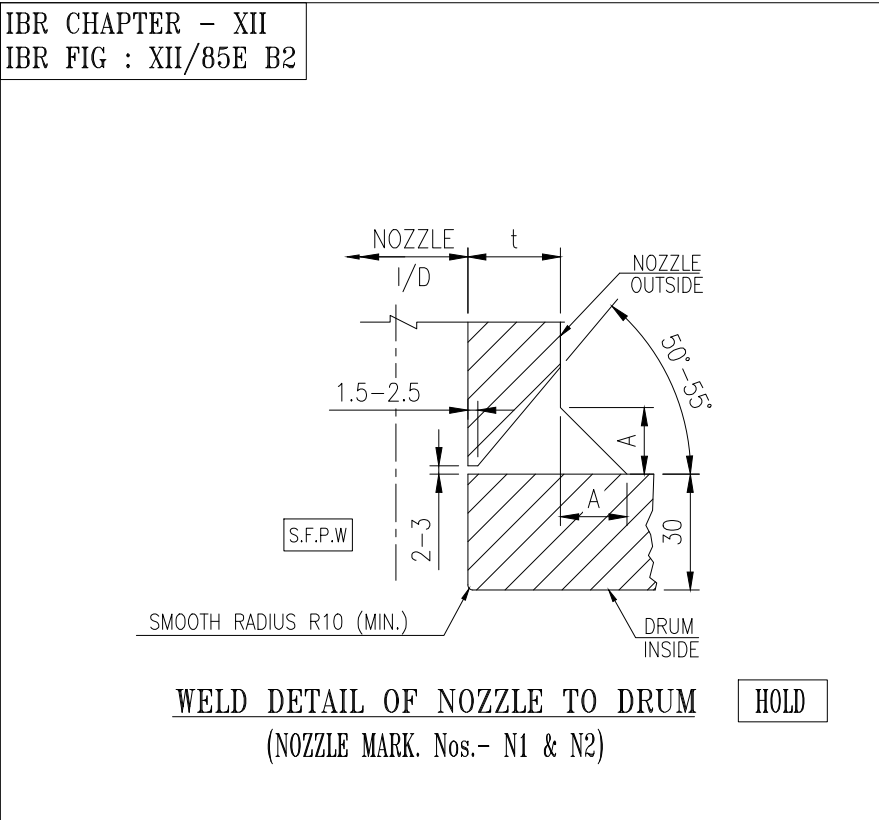
WELD PREPARATION OF CIRCUMFERENTIAL
SEAMS OF DRUM
FOR SEAM MK.Nos. CS-2, CS-3, CS-4 & CS-5



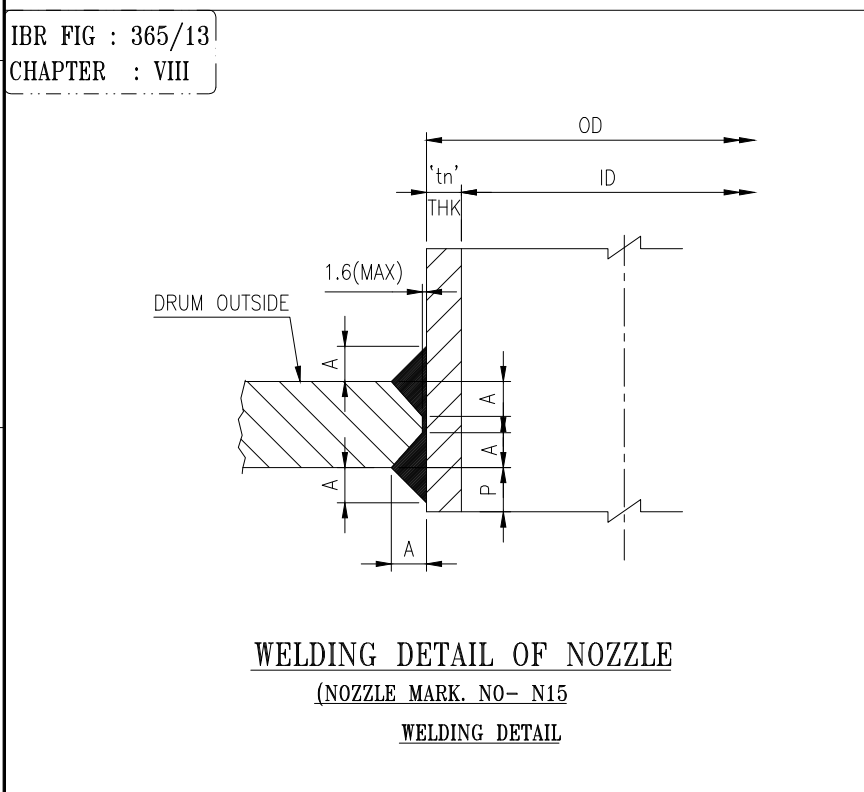
WELD PREPARATION OF CLOSING
SEAM OF DRUM
FOR SEAM MK.Nos. CS-1 & CS-6



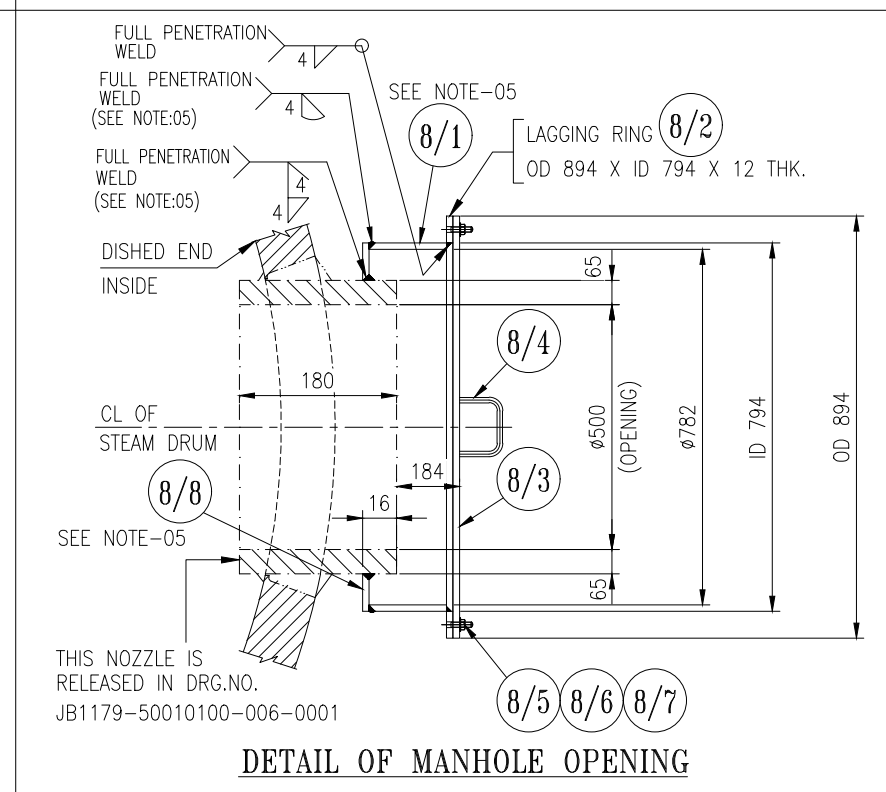
WELD DETAIL OF NOZZLE TO DRUM
(NOZZLE MARK Nos. N4, N9, N18, N20, N21
N26, N27, N31, N32, N38/1, N38/2 & N40)



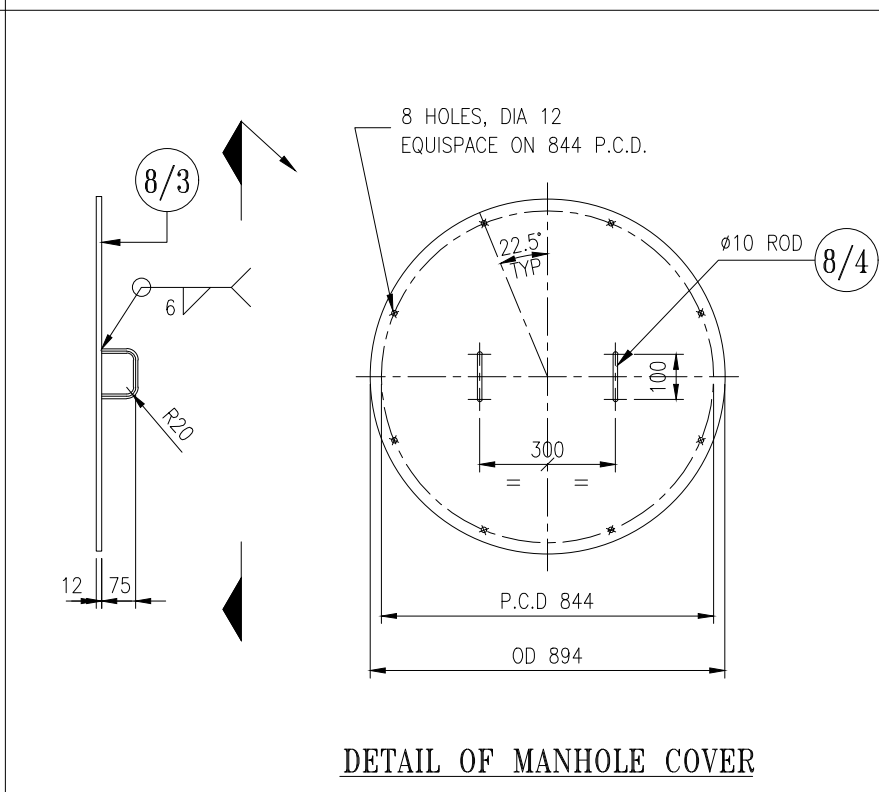
WELD DETAIL OF NOZZLE TO DRUM
(NOZZLE MARK Nos. - N1 & N2)



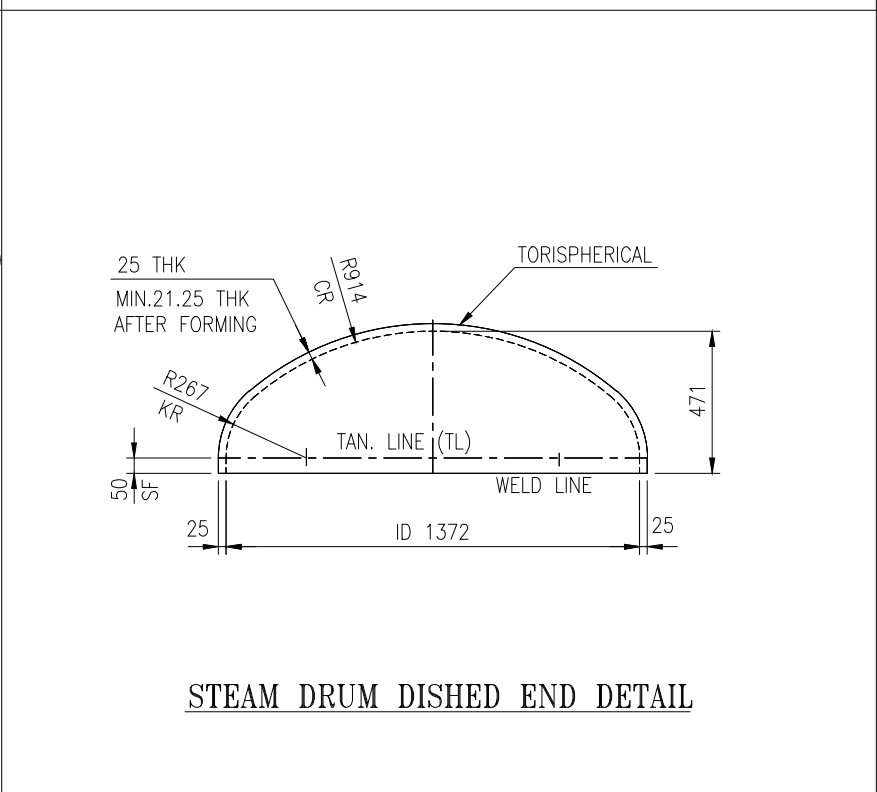
WELDING DETAIL OF NOZZLE
(NOZZLE MARK NO- N15
WELDING DETAIL



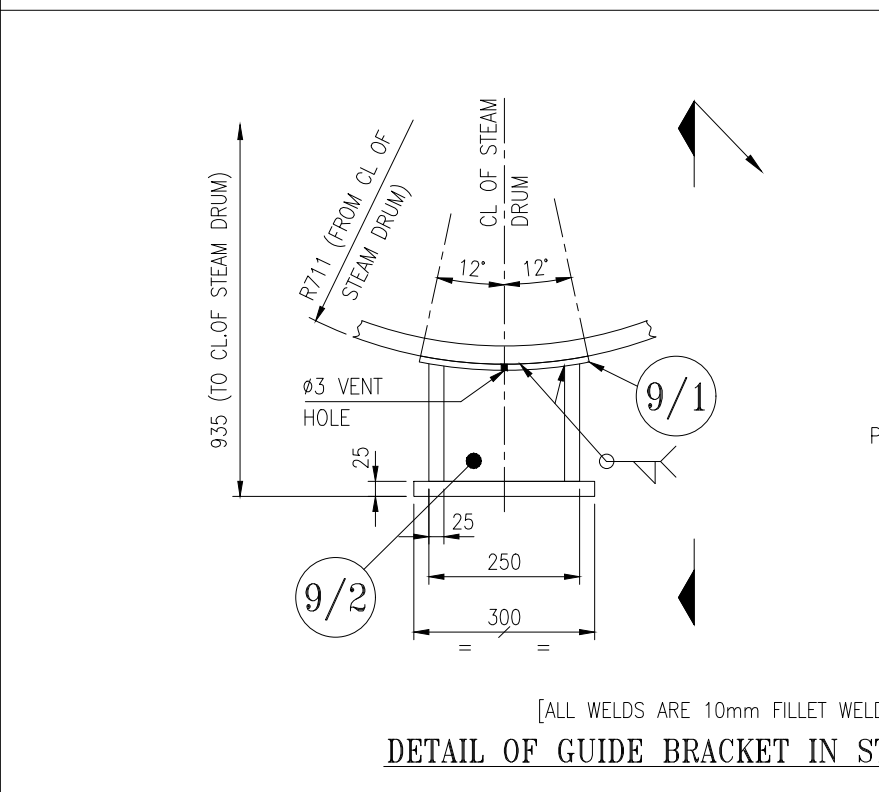
DETAIL OF MANHOLE OPENING



DETAIL OF MANHOLE COVER



STEAM DRUM DISHED END DETAIL



DETAIL OF GUIDE BRACKET IN STEAM DRUM

DESIGN DATA	
CODE OF CONSTRUCTION	I.B.R. (1950) AND IT'S AMMENDMENTS UPTO AUGUST 2020 (SEE NOTE:01)
MAXIMUM WORKING PRESSURE	26.5 Kg/Sq. cm (g)
MAXIMUM WORKING TEMPERATURE	228°C
WORKING METAL TEMPERATURE	256°C [INSIDE GAS PATH]
HYDRO TEST PRESSURE	39.75 Kg/Sq. cm (g) AS PER IBR AT SHOP & SITE
RADIOGRAPHY OF LONG & CIRCUM SEAM	100%
STRESS RELIEVING	YES REFER HEAT TREATMENT CYCLE TABLE
EVAPORATION OF BOILER (M.C.R.)	110 TPH
TOTAL HEATING SURFACE AREA OF BOILER (EXCLUDING ECONOMISER & SUPER HEATER)	1380 Sq.m (APPROX.)
HEATING SURFACE AREA OF ECONOMISER (NON STEAMING)	NOT APPLICABLE
HEATING SURFACE AREA OF DRUMS	25 Sq.m (APPROX.)
FLANGE PARTICULAR UNLESS STATED OTHERWISE	REFER DRG. NO.- JB1179-50010100-009-0001.
STEAM PRESSURE AT SUPER HEATER OUTLET	19.00 Kg/Sq. cm (g)
SUPER HEATER OUTLET TEMPERATURE	310±5°C
WEIGHT OF DRUMS	13820.94 Kgs (EMPTY)
INSPECTION	CIB (HARYANA) SEE NOTE: 02

S. No.	Nozzle Nos.	Nozzle Descriptions	Location of Nozzle	No. of Nozzle	Nozzle Size (OD) (mm)	Thickness of Nozzle	Nozzle Thickness (t) (mm)	Connecting Pipe Schedule Thickness (mm)	Projection from OD of Drum (mm)	Nozzle CL to Drum Horizontal Axis (mm)	Nozzle CL to Drum Vertical Axis (mm)	Overall Length (mm)	Finished Length (mm)	M/C Bar for end Preparation (mm)	Weld Dim. (mm)	Pg. in form of nozzle (mm)
1	N1	SAFETY VALVE NOZZLE-1	STEAM DRUM SHELL	1												0
2	N2	SAFETY VALVE NOZZLE-2	STEAM DRUM SHELL	1												0
3	N4	SATURATED STEAM STUB	STEAM DRUM SHELL	1	323.8	SCH.80	17.48	7.1 THK.	200			245	213	309.6	17	0
4	N9	THERMAL SLEEVE FOR FEED INLET NOZZLE	STEAM DRUM SHELL	1	219.1	SCH. 120	18.26						REF.DRG: -009-0001			125
5	N15	CBD NOZZLE	STEAM DRUM DISH	1	48.3	SCH.XXS	10.15	SCH.40			500	480	450	40.94	10	142
6	N18	AIR VENT NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	200			230	200	26.64	10	0
7	N20	INSTRUMENT TEST GAUGE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	200			230	200	26.64	10	0
8	N21	NITROGEN BLANKETING NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	200			230	200	26.64	10	0
9	N26	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM SHELL	1	48.3	SCH.XXS	10.15	SCH.40			460	420	389	40.94	11	0
10	N27	LEVEL GAUGE NOZZLE (WATER)	STEAM DRUM SHELL	1	48.3	SCH.XXS	10.15	SCH.40			460	420	389	40.94	11	0
11	N31	LEVEL TRANSMITTER (STEAM)	STEAM DRUM SHELL	1	48.3	SCH.XXS	10.15	SCH.40			305	310	280	40.94	11	0
12	N32	LEVEL TRANSMITTER (WATER)	STEAM DRUM SHELL	1	48.3	SCH.XXS	10.15	SCH.40			305	310	280	40.94	11	0
13	N38/1&2	PRESSURE GAUGE NOZZLE (LOCAL)	STEAM DRUM SHELL	2	33.4	SCH.XXS	9.09	SCH.40	200			230	200	26.64	10	0
14	N40	PRESSURE TRANSMITTER NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	200			230	200	26.64	10	0

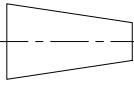
REV.	ZONE	DESCRIPTION	REV.	ZONE	DESCRIPTION	REV.	ZONE	DESCRIPTION	REV.	ZONE	DESCRIPTION	REV.	ZONE	DESCRIPTION	REV.	ZONE	DESCRIPTION

NOTES :-

01. THE ZERO DATE OF THIS CONTRACT IS 01.08.2020 HENCE IBR AMENDMENTS UPTO AUGUST 2020 HAVE BEEN TAKEN CARE OF WHILE DESIGNING THE EQUIPMENT.
02. INSPECTION BY CIB HARYANA BECAUSE MANUFACTURING OF THIS EQUIPMENT SHALL BE DONE AT ISGEC WORKS, YAMUNA NAGAR (HARYANA)
03. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE SPECIFIED.
04. FOR GENERAL NOTES & MATERIAL SPECIFICATION REF. DOCUMENT ID : JB1179-50010100-007-0002.
05. P. No. 8/8 RING PLATE FIRST TO BE WELDED WITH MANHOLE NOZZLE AFTER P. No. 8/1 (CLEAT FOR RING) WELDED WITH P. No. 8/8 (RING PLATE).

MATERIAL SPECIFICATIONS :					
DESCRIPTION	SIZE	SPECIFICATION	MIN. TENSILITY (Kg/ Sq.cm)	MIN.CALC. THK. (mm)	REMARKS
SHELL BELTS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	28.63	
DISHED ENDS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	18.86	

LIST OF REFERENCE DRAWINGS	
DRAWING DOCUMENT ID	DESCRIPTION
JB1179-50010100-006-0001	ARRANGEMENT & DETAIL OF CIRCULAR MANHOLE ON STEAM DRUM
JB1179-50010100-007-0002	DETAIL AND MATERIAL LIST OF STEAM DRUM
JB1179-50010100-008-0001	LAYOUT OF TUBE HOLES ON STEAM DRUM
JB1179-50010100-009-0001	ARRANGEMENT & DETAIL OF STEAM DRUM NOZZLES

Issued to _____			Issued by _____			Issued date _____		
☐	For pre-planning			☐	For manufacture in shop _____			
☐	For provisional manufacture in shop _____			☐	For sub contracting / site work			
This drawing supersedes drawing no. _____				Rev. _____				
W.O. No. PD - 0354						/ I.W.T. No. : 6678		
CONSULTANT		NTPC						
CLIENT		L&T MHPs BOILERS PVT LTD, BUXAR, BIHAR.						
DO NOT SCALE IF IN DOUBT PLEASE ASK				THIS DESIGN IS THE PROPERTY OF M/S. ISGEC HEAVY ENGINEERING LTD. AND IS ONLY ALLOWED TO BE USED BY EXPRESSED PERMISSION AND LICENSE FROM M/S. ISGEC HEAVY ENGINEERING LTD.				
FOR APPROVAL	NO	FOR INFORMATION	NO	FOR PRODUCTION	YES	FOR ERECTION	YES	
RESPONSIBLE DEPT.	DRAWN BY		CHECKED BY		APPROVED BY		DATE	
MECHANICAL	V.KAVASKAR		R.G.AMIR/R.KUMARAN		R.ARUNKUMAR		16.10.2020	
		TOTAL WEIGHT in kg.		PROJECT ID		ELEMENT ID		I.W.T. No.
		REF.007-0002		JB1179		50010100		6678
				YEAR : 2020		SHEET SIZE : A1		SCALE : 1 : 25
 ISGEC BOILERS (formerly Isgco John Thompson) Isgec Heavy Engineering Ltd. Noida 201 301 (U.P.) India				OIL FIRED AUXILIARY BOILER				
				SERVICE	ARRANGEMENT AND DETAIL OF STEAM DRUM			
				TITLE				
				DRAWING DOCUMENT ID				
				JB1179-50010100-007-0001		REVISION		00